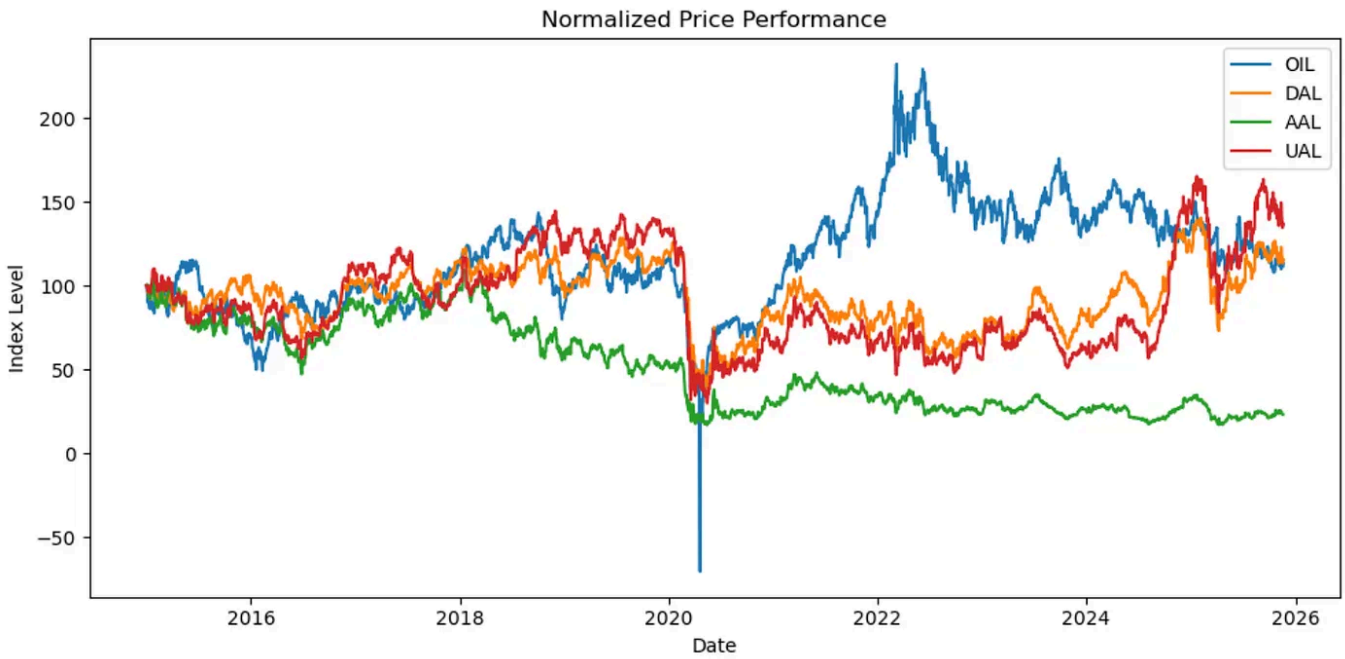
Open in app ↗ Sign up Sign in

Medium Search Write

Oil Prices vs Airline Stocks: The Relationship Everyone Gets Wrong

Aekank Patel Follow 4 min read · Dec 24, 2025

19



Normalized Price Performance of Oil and Major U.S. Airline Stocks

Airlines burn fuel.

Fuel comes from oil.

So it feels obvious that when oil prices rise, airline stocks should fall.

This idea is repeated so often that it sounds like a fact.

But when I tested it using **real Bloomberg data**, the result was not what I expected.

Oil prices alone explain almost nothing about airline stock returns.

The relationship only becomes clear when you look at the market properly.

The question I wanted to answer

I kept the question simple:

Do changes in oil prices actually move airline stocks?

To test this, I analyzed:

- Three major U.S. airlines: Delta (DAL), American (AAL), United (UAL)
- WTI crude oil futures
- The S&P 500 (SPX) as a market benchmark

Bloomberg Terminal was used for data extraction and exploratory analysis.

Python was used only where Bloomberg's tools had limitations.

First impression: airlines don't behave like oil

When you compare long-term price movements, oil and airline stocks behave very differently.

Oil prices swing wildly due to geopolitics, supply shocks, and global events.

Airline stocks, on the other hand, move together and follow broader equity trends.

Even visually, airlines look like **market-driven stocks**, not oil-linked assets.

Correlations confirm the intuition

1) Edit | 2) Actions | 3) Add Matrix Shortcut | 4) Settings | My Correlation Matrices: Correlation Matrix

01/01/2015 - 11/19/2025 | Weekly | Calculation Correlation | Local CCY

<Filter> | Correlation Matrix (4 Rows x 4 Columns)

Security	CL1	DAL	AAL	UAL
11) CL1	1.000	0.118	0.066	0.066
12) DAL	0.118	1.000	0.810	0.864
13) AAL	0.066	0.810	1.000	0.845
14) UAL	0.066	0.864	0.845	1.000

(Color Bands Based on Statistical Significance: Most Significant, Significant, Less Significant, Least Significant)
Click on a security to view more options, or on a matrix value to view correlation and scatter charts.

Bloomberg weekly return correlations.

Bloomberg's correlation matrix makes this clearer.

Oil's correlation with airline stocks is extremely low.

Meanwhile, airline stocks are highly correlated with each other.

In plain terms:

- Airlines move together
- Oil moves on its own

At this point, oil already looks like a weak driver.

Simple regressions: oil explains almost nothing

Bloomberg Regression (AAL on CL1, DAL on CL1 and UAL on CL1)

Next, I ran single-factor regressions using Bloomberg's Historical Regression Analysis (HRA).

Each airline's returns were regressed only on oil returns.

Get Aekank Patel's stories in your inbox

Join Medium for free to get updates from this writer.

Enter your email

Subscribe

Remember me for faster sign in

The results were striking:

- R^2 values below 1%
- Oil coefficients were small and statistically weak

Based on this, many people would stop here and say: *"Oil doesn't matter for airline stocks."*

But that conclusion misses something important.

The mistake: ignoring the market

Airline stocks are still stocks.

They rise and fall with:

- Market rallies
- Market crashes
- Investor sentiment

If the entire market goes up, airline stocks often go up too — even if oil prices rise at the same time.

That market effect can completely hide oil's impact.

So I asked a better question:

What happens if we control for the market?

Adding the market changes the story

Estimated Regression Statistics:

	alpha	beta_oil	t_beta_oil	beta_spx	t_beta_spx	r2	adj_r2
DAL	-0.000514	-0.043271	-2.961477	1.335777	36.213803	0.330753	0.330264
AAL	-0.001149	-0.087188	-4.578295	1.463709	30.445810	0.255237	0.254692
UAL	-0.000528	-0.040424	-2.177331	1.496790	31.935861	0.278623	0.278095

Multiple Regression Statistics

Bloomberg's regression tools allow only one variable at a time, so I extended the analysis in Python to include both:

- Oil returns
- Market returns (SPX)

Once the market was included, the results flipped.

Oil's effect became **negative and statistically significant** for all three airlines.

At the same time, the market factor remained large and dominant.

The model's explanatory power jumped from nearly zero to **about 25–33%**.

This is exactly what economic theory predicts.

Why this actually makes sense

Airlines are exposed to oil through fuel costs.

But they are far more exposed to the overall stock market.

Oil is not the main driver of airline stock returns.

It is a **secondary risk factor**.

You only see its effect clearly after separating out market noise.

The real takeaway

Oil prices alone don't meaningfully explain airline stock returns.

But once you control for the market, oil behaves exactly as theory suggests.

The broader lesson is simple but powerful:

Naive correlations and one-factor models can be misleading.

Sometimes the relationship only appears when you model the system properly.

Final thought

If an analysis feels obvious, it's often incomplete.

The most interesting insights usually appear **after** you control for what everyone else ignores.

Bloomberg | Airlines | Stocks | Finance | Data Science

Written by Aekank Patel

0 followers · 1 following

Follow

No responses yet

Write a response

What are your thoughts?

Recommended from Medium